

## Nuclear Astrophysics (NS 3035)

### Lec 5: Charged Particle Induced Reactions

The large blue spherical regions represent two positively charged nuclei separated by a Coulomb barrier.

Classically, a charged particle approaching another positively charged nucleus would require enough kinetic energy to overcome the electrostatic repulsion between them. At the temperatures found in stellar interiors, however, most particles do not possess sufficient energy to cross this barrier. The dark region between the nuclei symbolizes the potential barrier created by the Coulomb force. According to classical physics, particles with energies lower than the barrier height should be completely reflected. The colored beam passing through the barrier represents the quantum mechanical wave associated with the incoming charged particle. Unlike a classical particle, a quantum wavefunction does not abruptly vanish at the barrier. Instead, it penetrates into the classically forbidden region and retains a finite amplitude on the opposite side. Consequently, there exists a non-zero probability that the particle will appear beyond the barrier even when its energy is insufficient to overcome it classically.

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Charged Particle Induced Reactions

Astrophysical S-Factor and Gamow Peak

Gamow Peak Width, Temperature Dependence of Reaction Rates, and Electron Screening

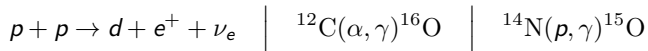
# Charged Particle Induced Reactions in Stars

Charged particle induced reactions constitute one of the most important classes of nuclear reactions in stellar environments. These reactions are responsible for both the production of stellar energy and the synthesis of new nuclei. Typical projectiles involved in these reactions are protons ( $p$ ), alpha particles ( $\alpha$ ), and occasionally heavier charged ions.

The charged particle reactions

- ▶ Power stars during hydrogen burning, helium burning, and later evolutionary stages
- ▶ Determine the rates of nucleosynthesis responsible for the origin of elements
- ▶ Their reaction rates strongly depend on temperature
- ▶ Their cross sections at stellar energies are extremely small and difficult to measure directly in laboratories
- ▶ Quantum tunneling plays a crucial role in allowing these reactions to occur inside stars

## Examples of Charged Particle Reactions



# Why Are Reaction Rates Important?

A star continuously evolves throughout its lifetime. As nuclear fuel is consumed, gravitational contraction and thermonuclear burning alter the internal temperature and density (**Recall first 3 classes**). For every stage of stellar evolution we must calculate reaction rates for thousands of nuclear reactions.

## Thermally Averaged Reaction Rate

$$\langle \sigma v \rangle = \left( \frac{8}{\pi \mu} \right)^{1/2} \frac{1}{(kT)^{3/2}} \int_0^{\infty} \sigma(E) E e^{-E/kT} dE$$

[See the notes for additional details and derivations](#)

### Meaning of terms

- ▶  $\sigma(E)$ : reaction cross section
- ▶  $v$ : relative velocity
- ▶  $\mu$ : reduced mass
- ▶  $T$ : stellar temperature

### This quantity determines

- ▶ Energy generation rates
- ▶ Evolutionary timescales
- ▶ Element production rates
- ▶ Final chemical composition

# The Challenge In Measuring $\sigma(E)$ for Stars

## Reaction Rates Depend on Cross Sections

Accurate calculation of stellar reaction rates requires full knowledge of the energy dependent nuclear cross section  $\sigma(E)$

## Why Is This a Difficult Problem?

Nuclear reactions inside stars occur at very low energies ( $\sim$  from a few keV to a few hundred keV)

- ▶ The Coulomb barrier strongly suppresses reaction probabilities
- ▶ Cross sections become extremely small
- ▶ Direct measurements in the laboratory are challenging
- ▶ Experimental signals are often buried beneath background radiation

# The Challenge In Measuring $\sigma(E)$ for Stars

## Reaction Rates Depend on Cross Sections

Accurate calculation of stellar reaction rates requires full knowledge of the energy dependent nuclear cross section  $\sigma(E)$

## How Do We Address This Challenge?

- ▶ Cross sections are measured at higher experimentally accessible energies
- ▶ Nuclear theory is used to model the energy dependence of  $\sigma(E)$
- ▶ The results are *extrapolated* to the lower energies relevant for stellar interiors

## The Central Goal

A major objective of nuclear astrophysics is to determine reliable low-energy cross sections and reaction rates for the conditions found inside stars.

# Non-Resonant and Resonant Reactions

The behavior of the reaction cross section determines the classification of nuclear reactions

## Non-Resonant Reactions

- ▶ Cross section varies smoothly with energy
- ▶ No sharp structures appear
- ▶ Easier to extrapolate to low energies
- ▶ Typical for many stellar fusion reactions



# Non-Resonant and Resonant Reactions

The behavior of the reaction cross section determines the classification of nuclear reactions

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## Resonant Reactions

- ▶ Cross section exhibits sharp peaks
- ▶ Strong enhancement occurs at specific energies
- ▶ Caused by formation of intermediate compound nuclear states
- ▶ Dominates reaction rates when resonance energies coincide with stellar energies

# More Questions

Several key questions arise when studying charged particle induced reactions

- ▶ Can the strong energy dependence of cross sections be removed?
- ▶ At what energies do most stellar reactions occur?
- ▶ Is the reaction probability the same across all energies?
- ▶ What energy window contributes most to stellar energy generation?
- ▶ How can we identify the most effective reaction energy?
- ▶ How does the reaction rate depend on temperature?

Answering these questions leads directly to the concepts of

- ▶ Coulomb barrier
- ▶ Quantum tunneling
- ▶ Astrophysical S-factor
- ▶ Gamow peak
- ▶ Temperature dependent reaction rates

# The Coulomb Barrier

Before two nuclei can fuse they must approach closely enough for the **strong nuclear force** to become effective. But, positively charged nuclei repel each other through the **Coulomb force**

$$V_C(r) = \frac{Z_1 Z_2 e^2}{r}$$

where  $Z_1$  = **projectile charge**,  $Z_2$  = **target charge** and  $r$  = **separation distance**

## Characteristics

- ▶ **Long-range** repulsive interaction.
- ▶ Produces an **energy barrier**.
- ▶ Prevents fusion at **low energies**.

## Consequences

- ▶ Fusion probability decreases rapidly.
- ▶ Barrier becomes larger for **heavier nuclei**.
- ▶ Higher kinetic energy is required for fusion.

Only when nuclei approach within the **nuclear radius** does the attractive **strong nuclear force** dominate over the **Coulomb repulsion**, allowing fusion to occur.

# Effective Nuclear Potential

The interaction between two nuclei is governed by two competing forces.

## Repulsive Interaction

### Coulomb Force

- ▶ Acts over long distances.
- ▶ Repels positively charged nuclei.
- ▶ Produces the Coulomb barrier.

## Attractive Interaction

### Strong Nuclear Force

- ▶ Extremely strong.
- ▶ Short-range interaction.
- ▶ Binds nucleons together.

# Effective Nuclear Potential

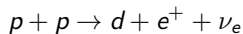
## For Fusion to Occur

- ▶ Classically: Projectile energy must exceed the barrier height.
- ▶ Quantum mechanically: nuclei can penetrate the barrier even when  $E < V_C$ .

**This phenomenon is called Quantum Tunneling**

# The Proton-Proton Fusion Problem

The most important reaction in stellar astrophysics is



For two protons  $V_C \approx 550$  keV. The corresponding temperature required for classical barrier crossing is  $T \approx 6.4 \times 10^9$  K or  $T_9 \approx 6.4$ . This leads to a paradox...

- ▶ If stars required such temperatures, proton fusion would occur explosively
- ▶ Stars would burn their fuel almost instantaneously
- ▶ Stable stars like the Sun could not exist

Yet the Sun burns steadily for billions of years. So clearly another mechanism must allow fusion at much lower temperatures

# Why Classical Physics Fails

Consider a typical stellar core temperature

$$T = 10^7 \text{ K}$$

or

$$T_9 = 0.01$$

The thermal energy is

$$kT \approx 0.86 \text{ keV}$$

Compare this with the proton-proton Coulomb barrier:  $V_C \approx 550 \text{ keV}$

Thus:  $kT \ll V_C$  The average proton possesses nearly 600 times less energy than required for classical fusion. Classical mechanics predicts essentially zero fusion probability. Therefore classical physics alone cannot explain stellar energy production.

# Maxwell-Boltzmann Distribution is Not Enough

One possible argument is that a few particles in the high-energy tail of the Maxwell-Boltzmann distribution might possess sufficient energy to overcome the barrier

However, the probability ratio is approximately

$$\frac{P(550 \text{ keV})}{P(0.86 \text{ keV})} \approx 10^{-275}$$

This means

- ▶ Particles capable of classical fusion are essentially absent
- ▶ Thermal fluctuations alone cannot explain stellar luminosities
- ▶ Another mechanism must be operating

The observed energy output of stars therefore provides direct evidence for a non-classical process occurring in stellar interiors



# Gamow's Solution → Quantum Tunneling

## The Classical Problem

Charged particles inside stars possess energies far below the Coulomb barrier. Classically, fusion should therefore be impossible.

## Quantum Mechanical Solution

George Gamow proposed that

- ▶ Particles behave as matter waves
- ▶ The wavefunction extends into the forbidden region
- ▶ A particle can penetrate the Coulomb barrier even when  $E < V_C$
- ▶ This process is called **quantum tunneling**
- ▶ Once inside the nuclear radius, the strong force can cause fusion

Quantum tunneling allows nuclear reactions to occur at stellar temperatures far below those required for classical barrier penetration

# Gamow's Solution → Quantum Tunneling

## Physical Picture

- ▶ A particle does **not** need energy greater than the barrier height
- ▶ The quantum wavefunction extends into the classically forbidden region
- ▶ A finite probability exists for the particle to appear on the other side of the barrier.
- ▶ Once inside the nuclear radius, the strong force can trigger fusion

$$P_{\text{tunnel}} \propto e^{-2G}$$

where  $G$  is the **Gamow factor**

## Typical Tunneling Probabilities

$$E = 1 \text{ keV} \Rightarrow P \sim 10^{-10}$$

$$E = 2 \text{ keV} \Rightarrow P \sim 10^{-7}$$

$$E = 5 \text{ keV} \Rightarrow P \sim 10^{-4}$$

## Remark

Even these tiny probabilities are enormously larger than the classical prediction

$$P_{\text{classical}} \approx 10^{-275}$$

# Importance of Quantum Tunneling in Stars

## Without Tunneling

- ▶ Fusion requires  $T \sim 10^9$  K
- ▶ Extremely high reaction rates
- ▶ Stars would burn fuel rapidly
- ▶ Long stellar lifetimes impossible
- ▶ Stable stars could not exist

## With Tunneling

- ▶ Fusion occurs near  $T \sim 10^7$  K
- ▶ Controlled energy generation
- ▶ Gradual fuel consumption
- ▶ Lifetimes of billions of years
- ▶ Continuous nucleosynthesis

## Take Home Message

The existence of stars, planets, heavy elements, and ultimately life depends on the quantum mechanical tunneling of charged particles through the Coulomb barrier.

# Summary

- ▶ Charged particle induced reactions are the primary energy source in stars
- ▶ Reaction rates depend strongly on the nuclear cross section and stellar temperature
- ▶ Non-resonant reactions exhibit smooth energy dependence of the cross section
- ▶ Positively charged nuclei face a Coulomb barrier that suppresses fusion
- ▶ Proton-proton fusion requires approximately 550 keV classically
- ▶ Typical stellar temperatures provide only  $\sim 1$  keV thermal energies
- ▶ Maxwell-Boltzmann statistics alone cannot explain stellar fusion
- ▶ Quantum tunneling provides a finite barrier penetration probability
- ▶ Gamow's theory explains how fusion occurs at stellar temperatures

**The existence of stars is one of the most spectacular manifestations of quantum mechanics on a cosmic scale.**

# Outline

Charged Particle Induced Reactions

Astrophysical S-Factor and Gamow Peak

Gamow Peak Width, Temperature Dependence of Reaction Rates, and Electron Screening

# The Need for an Astrophysical S-Factor

For charged particle induced reactions, the nuclear reaction cross section exhibits a very strong dependence on the incident energy. This dependence arises primarily from the Coulomb repulsion between the interacting nuclei. As the energy decreases below the Coulomb barrier, the probability of penetrating the barrier decreases exponentially, causing the cross section to become extremely small

In stellar environments, most reactions occur at energies of only a few keV to a few hundred keV, far below the Coulomb barrier. Hence

- ▶ Cross sections become exceedingly small
- ▶ Direct laboratory measurements become difficult
- ▶ Experimental uncertainties increase rapidly
- ▶ Stellar reaction rates often require extrapolation from higher-energy measurements

A major challenge in nuclear astrophysics is therefore to separate the rapidly varying Coulomb barrier effects from the more slowly varying nuclear structure effects. This motivates the introduction of the **Astrophysical S-factor**, one of the most important quantities in stellar reaction theory.

# Quantum Tunneling and the Gamow Factor

For charged-particle reactions in stars, the projectile energy is usually much smaller than the Coulomb barrier  $E \ll V_C$

Classically, fusion is impossible because the particles do not have enough energy to overcome the electrostatic repulsion

Quantum mechanics allows the particles to tunnel through the classically forbidden region. To estimate the tunneling probability we make the following assumptions:

- ▶ Relative motion can be treated as a two-body problem with reduced mass  $\mu$
- ▶ Outside the nuclear radius, the interaction is purely Coulomb:  $V(r) = \frac{Z_1 Z_2 e^2}{r}$ .
- ▶ The particle energy satisfies  $E < V(r)$  in the forbidden region.
- ▶ The WKB approximation is valid because the wavefunction varies slowly compared with its wavelength.
- ▶ Nuclear forces act only after the Coulomb barrier has been penetrated.

The resulting tunneling probability is

$$P(E) \propto e^{-2\pi\eta}$$

# Quantum Tunneling and the Gamow Factor

The resulting tunneling probability is

$$P(E) \propto e^{-2\pi\eta}$$

where  $\eta = \frac{Z_1 Z_2 e^2}{\hbar v}$  is called the **Sommerfeld parameter**.

Using  $v = \sqrt{\frac{2E}{\mu}}$  gives

$$2\pi\eta = 31.29 Z_1 Z_2 \sqrt{\frac{\mu}{E}},$$

with  $E$  in keV and  $\mu$  in amu.



# Quantum Tunneling and the Gamow Factor

Starting from the radial Schrödinger equation

$$-\frac{\hbar^2}{2\mu} \frac{d^2 u}{dr^2} + V(r)u = Eu,$$

the classically forbidden region satisfies  $V(r) > E$ .

In the WKB approximation, the transmission probability through a barrier is

$$P(E) \simeq \exp \left[ -2 \int_{r_N}^{r_c} \kappa(r) dr \right],$$

where

$$\kappa(r) = \frac{\sqrt{2\mu [V(r) - E]}}{\hbar}.$$

# Quantum Tunneling and the Gamow Factor

For a Coulomb potential,

$$V(r) = \frac{Z_1 Z_2 e^2}{r},$$

the outer classical turning point is obtained from  $V(r_c) = E$ ,

$$r_c = \frac{Z_1 Z_2 e^2}{E}.$$

Since the nuclear radius satisfies  $r_N \ll r_c$ ,  
the WKB integral becomes

$$\int_{r_N}^{r_c} \kappa(r) dr \approx \pi\eta.$$

Therefore,

$$P(E) = e^{-2\pi\eta}.$$

# Quantum Tunneling and the Gamow Factor

Defining the Sommerfeld parameter as

$$\eta = \frac{Z_1 Z_2 e^2}{\hbar v},$$

we obtain the famous **Gamow factor**

$$G(E) = e^{-2\pi\eta}.$$

Using  $v = \sqrt{2E/\mu}$ ,

$$2\pi\eta = 31.29 Z_1 Z_2 \sqrt{\frac{\mu}{E}},$$

where  $E$  is in keV and  $\mu$  is in amu.

The Gamow factor represents the exponential suppression of low-energy charged-particle fusion reactions due to Coulomb barrier penetration.

# Cross Section for Non-Resonant Reactions

For a non-resonant charged particle induced reaction, the energy dependence of the cross section can be written as

$$\sigma(E) = \frac{S(E)}{E} e^{-2\pi\eta}.$$

This expression separates the physics into three distinct contributions:

1.  **$1/E$  term**

Arises primarily from the de Broglie wavelength dependence of the reaction probability.

2.  **$e^{-2\pi\eta}$  term**

Represents Coulomb barrier penetration through quantum tunneling.

3.  **$S(E)$  term**

Contains the intrinsic nuclear structure information and nuclear matrix elements.

Unlike the exponential Gamow factor, the S-factor varies slowly with energy for non-resonant reactions. This slow variation makes it considerably easier to extrapolate measured data toward astrophysical energies.

# Definition and Physical Meaning of the S-Factor

The Astrophysical S-factor is defined as

$$S(E) = \sigma(E) E e^{2\pi\eta}.$$

Physically, the S-factor removes the two dominant sources of energy dependence:

- ▶ The geometrical  $1/E$  dependence.
- ▶ The Coulomb tunneling dependence  $e^{-2\pi\eta}$ .

As a result, the remaining quantity reflects primarily:

- ▶ Nuclear structure effects.
- ▶ Spectroscopic properties.
- ▶ Overlap between initial and final nuclear wave functions.
- ▶ Transition matrix elements.

For non-resonant reactions,  $S(E)$  is usually a slowly varying function and may often be approximated by a low-order polynomial expansion:

# Why the S-Factor Improves Extrapolation

Experimental measurements are rarely performed exactly at stellar energies.

For example:

- ▶ Typical stellar fusion energies may lie between 10 and 300 keV.
- ▶ Experimental measurements are often performed at significantly higher energies.
- ▶ Direct measurements become increasingly difficult as cross sections fall exponentially.

If one attempts to extrapolate the cross section directly,

$$\sigma(E),$$

small uncertainties in energy can generate enormous uncertainties in the extrapolated result because of the exponential dependence.

However, after removing the Coulomb suppression,

$$S(E) = \sigma(E) E e^{2\pi\eta},$$

the remaining quantity varies much more smoothly.

Therefore:

# Reaction Rate in Terms of the S-Factor

The thermally averaged reaction rate per particle pair is

$$\langle \sigma v \rangle = \left( \frac{8}{\pi \mu} \right)^{1/2} \frac{1}{(kT)^{3/2}} \int_0^\infty E \sigma(E) e^{-E/kT} dE.$$

Substituting

$$\sigma(E) = \frac{S(E)}{E} e^{-2\pi\eta},$$

gives

$$\langle \sigma v \rangle = \left( \frac{8}{\pi \mu} \right)^{1/2} \frac{1}{(kT)^{3/2}} \int_0^\infty S(E) e^{-2\pi\eta} e^{-E/kT} dE.$$

The reaction rate therefore depends on the product of two competing exponentials:

$$e^{-2\pi\eta}$$

and

# Origin of the Gamow Peak

Two opposing physical effects determine the probability that a fusion reaction occurs.

## Effect 1: Maxwell-Boltzmann Distribution

$$e^{-E/kT}$$

favours low energies because high-energy particles are relatively rare.

## Effect 2: Tunneling Probability

$$e^{-2\pi\eta}$$

favours higher energies because penetration through the Coulomb barrier becomes easier.

These two effects compete:

- ▶ Low energies contain many particles but poor tunneling probability.
- ▶ High energies tunnel efficiently but contain very few particles.

The product

$$e^{-E/kT} e^{-2\pi\eta}$$



# Physical Interpretation of the Gamow Peak

The Gamow Peak answers a fundamental question:

*At what energies do most stellar fusion reactions occur?*

The answer is not:

$$E = kT.$$

Although the Maxwell-Boltzmann distribution peaks at  $kT$ , fusion reactions occur preferentially at energies higher than  $kT$  because tunneling becomes more effective there.

The centroid of the Gamow Peak is denoted by

$$E_0.$$

Typically

$$E_0 > kT.$$

Thus stellar reactions occur mainly in the high-energy tail of the Maxwell-Boltzmann distribution rather than at the thermal peak itself

# Gamow Peak Energy

The most effective energy for charged-particle fusion is obtained by maximizing the integrand of the reaction-rate expression.

The result is

$$E_0 = 1.22 (Z_1^2 Z_2^2 \mu T_6^2)^{1/3} \text{ keV},$$

where

- ▶  $Z_1, Z_2$  are nuclear charges,
- ▶  $\mu$  is the reduced mass in amu,
- ▶  $T_6$  is temperature in units of  $10^6$  K.

Important conclusions:

- ▶  $E_0$  increases with temperature.
- ▶  $E_0$  increases with nuclear charge.
- ▶ Heavier charged nuclei require higher effective reaction energies.
- ▶ Even for large nuclei,  $E_0$  remains below the Coulomb barrier.

# Examples of Gamow Peak Energies

Typical values of  $E_0$  include:

$$p + p \quad E_0 \approx 5.9 \text{ keV}$$

$$p + {}^{14}\text{N} \quad E_0 \approx 26.5 \text{ keV}$$

$$\alpha + \alpha \quad E_0 \approx 56 \text{ keV}$$

$${}^{12}\text{C} + \alpha \quad E_0 \approx 237 \text{ keV}$$

These examples illustrate a general trend:

$$E_0 \uparrow \quad \text{as} \quad Z_1 Z_2 \uparrow$$

because larger nuclear charges create stronger Coulomb barriers.

Nevertheless,

$$E_0 < V_C$$

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